



第50讲

不定积分概念、不定积分性质

在实际中, 经常要解决已知 $F'(x) = f(x)$
($f(x)$ 已知) $\Leftrightarrow f(x)dx = F'(x)dx = dF(x)$
求 $F(x)$

定义: 设 $f(x)$ 在区间 I 上有定义. 若存在一个
 $F(x)$ 对每一个 $x \in I$ 都有 $F'(x) = f(x)$, 称
 $F(x)$ 是 $f(x)$ 的一个原函数.

若 $F(x)$ 是 $f(x)$ 在区间 I 上的一个原函数,
则 $F(x) + C$ 也是 $f(x)$ 在区间 I 上的原函数
(C 为常数, $C \in \mathbb{R}$)

$\forall G(x)$ 是 $f(x)$ 在区间 I 上的任意一个原函数
即 $\forall x \in I$, 有 $G'(x) = f(x)$

由 $[G(x) - F(x)]' = G'(x) - F'(x) = f(x) - f(x) = 0$
知 $G(x) - F(x) = C$ (常) ($x \in I$)

即 $G(x) = F(x) + C$

定理: 若 $F(x)$ 是 $f(x)$ 在区间 I 上的一个原函数,

则 $F(x) + C$ (C 为常数, $C \in \mathbb{R}$) 是 $f(x)$ 在区间 I
上的全体原函数. 称为 $f(x)$ 在区间 I
上的不定积分. 记作 $\int f(x)dx$ 即

$\int f(x)dx = F(x) + C$ ($x \in I$, C 为常, $C \in \mathbb{R}$)

$f(x)$ 称为被积函数 $f(x)dx$ 称为被积

表达式. x 称为积分变量. " $\int$ " 称为不定积分号.

性质1: $(\int f(x)dx)' = \frac{d}{dx} \int f(x)dx = f(x)$

性质2: $d \int f(x)dx = f(x)dx$

性质3: $\int f'(x)dx = f(x) + C$

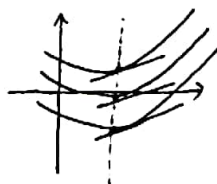
性质4: $\int d f(x) = f(x) + C$

即 $\int g(x)dx = \int dG(x) = G(x) + C$

不定积分的几何意义

设 $F(x)$ 是 $f(x)$ 的原函数. 则 $\int f(x)dx = F(x) + C$

知 $(\int f(x)dx)' = (F(x) + C)' = f(x)$



切线斜率相同.

基本的积分公式.

① $\int 0 dx = C$ ② $\int 1 dx = \int dx = x + C$

③ $\int x^a dx = \frac{1}{a+1} x^{a+1} + C$ ($a \neq -1$)

④ $\int x^{-1} dx = \int \frac{1}{x} dx = \ln|x| + C$

⑤ $\int a^x dx = \frac{a^x}{\ln a} + C$

⑥ $\int e^x dx = e^x + C$

⑦ $\int \cos x dx = \sin x + C$ ⑧ $\int \sin x dx = -\cos x + C$

⑨ $\int \sec^2 x dx = \tan x + C$

⑩ $\int \csc^2 x dx = -\cot x + C$





$$\textcircled{11} \int \sec x \tan x dx = \sec x + C$$

$$\textcircled{12} \int \csc x \cot x dx = -\csc x + C$$

$$\textcircled{13} \int \frac{1}{1+x^2} dx = \arctan x + C \\ = -\operatorname{arccot} x + C_1$$

$$\textcircled{14} \int \frac{1}{\sqrt{1-x^2}} dx = -\arccos x + C_1 \\ = \arcsin x + C$$

双曲余弦 $\cosh x = \frac{e^x + e^{-x}}{2}$

双曲正弦 $\sinh x = \frac{e^x - e^{-x}}{2}$

$$\cosh^2 x - \sinh^2 x = 1$$

$$(\sinh x)' = \cosh x \quad (\cosh x)' = \sinh x$$

$$\textcircled{15} \int \sinh x dx = \cosh x + C$$

$$\textcircled{16} \int \cosh x dx = \sinh x + C$$

不定积分的四大法宝.

一. 不定积分的线性运算法则

若 $\int f(x) dx$, $\int g(x) dx$ 均存在.

$\forall$ 常数 α, β (且 α, β 不同时为 0)

则 $\int [\alpha f(x) + \beta g(x)] dx$ 存在

$$= \alpha \int f(x) dx + \beta \int g(x) dx.$$

$$\text{证: } [\alpha \int f(x) dx + \beta \int g(x) dx]'$$

$$= \alpha [\int f(x) dx]' + \beta [\int g(x) dx]'$$

$$= \alpha f(x) + \beta g(x)$$

$\alpha \int f(x) dx + \beta \int g(x) dx$ 确实是

$\alpha f(x) + \beta g(x)$ 的原函数.

且含有常数 C , 故成立

$$\alpha \int f(x) dx + \beta \int g(x) dx = \alpha (F(x) + C_1) + \beta (G(x) + C_2) \\ = \alpha F(x) + \beta G(x) + C$$

例: $\int \sqrt{x} \sqrt{x} dx$

$$= \int (x(x^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}} dx$$

$$= \int x^{\frac{1}{2}} x^{\frac{1}{4}} x^{\frac{1}{8}} dx$$

$$= \int x^{\frac{7}{8}} dx$$

$$= \frac{8}{15} x^{\frac{15}{8}} + C$$

例: $\int \tan^2 x dx$

解: 原式 = $\int (\sec^2 x - 1) dx$

$$= \int \sec^2 x dx - \int 1 dx$$

$$= \tan x + C_1 - (x + C_2)$$

$$= \tan x - x + C$$

例: $\int \frac{1}{\sin^2 x \cos^2 x} dx$

解: 原式 = $\int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} dx$

$$= \int (\sec^2 x + \csc^2 x) dx$$

$$= \int \sec^2 x dx + \int \csc^2 x dx$$

$$= \tan x - \cot x + C$$





例: $\int \frac{x^4}{x^2+1} dx$

解: 原式 = $\int \frac{x^4+1-1}{x^2+1} dx$

= $\int (x^2-1+\frac{1}{x^2+1}) dx$

= $\frac{1}{3}x^3 - x + \arctan x + C$

或者 $\int g(x) dx = \int f(\varphi(x)) \varphi'(x) dx$
= $\int f(\varphi(x)) d\varphi(x)$

$\begin{aligned} &\stackrel{u=\varphi(x)}{=} \int f(u) du \\ &= \int dF(u) \\ &= \int dF(\varphi(x)) \\ &= F(\varphi(x)) + C \end{aligned}$

二、不定积分的凑微分(第一换元法)

$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = ?$

若 $F'(u) = f(u)$ 则 $[F(\varphi(x))]' = F'(\varphi(x)) \varphi'(x)$
= $f(\varphi(x)) \varphi'(x)$

求 $\int g(x) dx$. 如果 $g(x) dx = d?$ 则

$\int g(x) dx = ? + C$

如果 $g(x) dx = f(\varphi(x)) \varphi'(x) dx$

找到 $F(u)$ $f(\varphi(x)) d\varphi(x)$

$F(u) = f(u)$

令 $\varphi(x) = u$ $f(u) du = dF(u)$

$dF(u) = F'(u) du = f(u) du$

$\stackrel{u=\varphi(x)}{=} dF(\varphi(x))$

则 $\int g(x) dx = F(\varphi(x)) + C$

例: $\int e^{2x} dx$

$\varphi(x) = 2x$ $f(u) = e^u$

$f(\varphi(x)) = e^{2x}$

原式 = $\int e^{2x} (2x)' \cdot \frac{1}{2} dx$

= $\frac{1}{2} \int e^{2x} (2x)' dx$

$f(\varphi(x)) \varphi'(x)$

= $\frac{1}{2} \int e^{2x} d(2x)$

$\stackrel{u=2x}{=} \frac{1}{2} \int e^u du$

= $\frac{1}{2} e^u + C$

= $\frac{1}{2} e^{2x} + C$

例: $\int \tan x dx$

= $\int \frac{\sin x}{\cos x} dx$

= $-\int \frac{1}{\cos x} d \cos x$

= $-\ln|\cos x| + C$

⑪ $\int \tan x dx$

= $-\ln|\cos x| + C$

⑫ $\int \cot x dx$

= $\ln|\sin x| + C$

即 $\int g(x) dx = \int f(\varphi(x)) d\varphi(x) = F(\varphi(x)) + C$

记住一些微分关系式.

1. $dx = 1 \cdot dx = d \frac{1}{a} (ax+b) \quad (a \neq 0)$

即 $dx = \frac{1}{a} d(ax+b)$





$$2. x dx = \frac{1}{2} d(x^2 \pm a^2)$$

$$3. x dx = -\frac{1}{2} d(a^2 - x^2)$$

$$4. \cos x dx = d \sin x$$

$$5. \sin x dx = -d \cos x$$

$$6. \frac{1}{x} dx = d \ln|x|$$

$$\stackrel{x>0}{=} d \ln x$$

$$7. e^x dx = d e^x$$

若 $F'(u) = f(u)$

$$\int f(ax+b) dx = \frac{1}{a} \int f(ax+b) d(ax+b)$$

$$= \frac{1}{a} F(ax+b) + C$$

$$\int f(x) \int f(a^2 - x^2) x dx = -\frac{1}{2} \int f(a^2 - x^2) d(a^2 - x^2)$$

$$= -\frac{1}{2} F(a^2 - x^2) + C$$

$$\text{例: } \int \frac{1}{x^2 + a^2} dx \quad (a \neq 0)$$

$$= \frac{1}{a^2} \int \frac{1}{1 + (\frac{x}{a})^2} dx$$

$$= \frac{1}{a} \int \frac{1}{1 + (\frac{x}{a})^2} d(\frac{x}{a})$$

$$= \frac{1}{a} \arctan \frac{x}{a} + C$$

$$\textcircled{19} \int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$\text{例: } \int \frac{1}{\sqrt{a^2 - x^2}} dx \quad (a > 0)$$

$$= \frac{1}{a} \int \frac{1}{\sqrt{1 - (\frac{x}{a})^2}} dx$$

$$= \int \frac{1}{\sqrt{1 - (\frac{x}{a})^2}} d(\frac{x}{a})$$

$$= \arcsin \frac{x}{a} + C$$

$$\textcircled{20} \int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \frac{x}{a} + C$$

$$\text{例: } \int \frac{1}{a^2 - x^2} dx \quad (a \neq 0)$$

$$= \int \frac{1}{(a-x)(a+x)} dx$$

$$= \frac{1}{2a} \int (\frac{1}{a-x} + \frac{1}{a+x}) dx$$

$$= \frac{1}{2a} [\int \frac{1}{a-x} dx + \int \frac{1}{a+x} dx]$$

$$= \frac{1}{2a} [-\int \frac{1}{a-x} d(a-x) + \int \frac{1}{a+x} d(a+x)]$$

$$= -\frac{1}{2a} \ln|a-x| + \frac{1}{2a} \ln|a+x| + C$$

$$= \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\textcircled{21} \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\text{例: } \int \sec x dx$$

$$= \int \frac{1}{\cos x} dx$$

$$= \int \frac{\cos x}{\cos^2 x} dx$$

$$= \int \frac{1}{1 - \sin^2 x} d \sin x$$

$$\text{由 } \textcircled{21} \quad a=1 \quad x = \sin x \text{ 代入}$$

$$\text{原式} = \frac{1}{2} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| + C$$

$$= \frac{1}{2} \ln \left| \frac{(1 + \sin x)^2}{\cos^2 x} \right| + C$$

$$= \ln \left| \frac{1 + \sin x}{\cos x} \right| + C$$

$$= \ln |\sec x + \tan x| + C$$

$$\textcircled{22} \int \sec x dx$$

$$= \ln |\sec x + \tan x| + C$$

$$[\text{法}] \text{原式} = \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx$$

$$= \int \frac{1}{\sec x + \tan x} d(\tan x + \sec x)$$

$$= \ln |\sec x + \tan x| + C$$

同理,

$$\textcircled{23} \int \csc x dx = \ln |\csc x - \cot x| + C$$





$$\int e^{ax} dx \quad (a \neq 0)$$

$$= \frac{1}{a} \int e^{ax} d(ax)$$

$$= \frac{1}{a} e^{ax} + C$$

$$(24) \int e^{ax} dx = \frac{1}{a} e^{ax} + C \quad (a \neq 0)$$

$$(25) \int \cos ax dx = \frac{1}{a} \sin ax + C$$

$$(26) \int \sin ax dx = -\frac{1}{a} \cos ax + C$$

例 求 $\int \sqrt{a^2 - x^2} dx \quad (a > 0)$

三. 不定积分的变量代换

(第=换元法)

求 $\int f(x) dx$ $f(x) dx = d?$

由 $f(x) dx \xrightarrow{x=\varphi(t)} f(\varphi(t)) d\varphi(t)$

(要求 $\varphi(t)$ 可导) $= \boxed{f(\varphi(t)) \varphi'(t)} dt$
 $\underline{F(t) = f(\varphi(t)) \varphi'(t)} \quad dF(t)$

$x = \varphi(t)$ 严格单调
 $\underline{t = \varphi^{-1}(x)}$ $= dF(\varphi^{-1}(x))$

(t 用 x 表示, 并且一一对应)

$$\therefore \int f(x) dx = F(\varphi^{-1}(x)) + C$$

$$\int f(x) dx \xrightarrow[\text{严格单调}]{x=\varphi(t) \text{ 可导}} \int f(\varphi(t)) d\varphi(t)$$

$$= \int f(\varphi(t)) \varphi'(t) dt$$

$$\underline{F(t) = f(\varphi(t)) \varphi'(t)} \quad F(t) + C$$

$$= F(\varphi^{-1}(x)) + C$$

如果被积函数中含有下列根式,
且不能用前面方法(线性运算法则, 凑微分, 直接求)
此时可用变量代换 (去根式, 两项平方差 $\rightarrow$ 一项平方)

$$\sqrt{a^2 - x^2}, \text{ 令 } x = a \sin t \quad t \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\sqrt{a^2 + x^2}, \text{ 令 } x = a \tan t, \quad t \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\sqrt{x^2 - a^2}, \text{ 令 } x = a \sec t, \quad t \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$$

$$\sqrt{\frac{ax+b}{cx+d}}, \text{ 令 } \sqrt{\frac{ax+b}{cx+d}} = t \text{ 解出 } x = \varphi(t) \text{ 的有理式.}$$

$$\text{令 } x = \varphi(t).$$

特别地, $\sqrt{ax+b} = t$

例 $\int \sqrt{a^2 - x^2} dx \quad (a > 0)$

解: 原式 $\xrightarrow[t \in [-\frac{\pi}{2}, \frac{\pi}{2}]]{x = a \sin t} \int \sqrt{a^2 - a^2 \sin^2 t} d(a \sin t)$

$$= a \int |\cos t| a \cos t dt$$

$$= a^2 \int \cos^2 t dt$$

$$= \frac{a^2}{2} \int (1 + \cos 2t) dt$$

$$= \frac{a^2}{2} t + \frac{a^2}{4} \sin 2t + C$$

$$= \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{a^2}{2} \sin t \cos t + C$$

$$\begin{aligned} x &= a \sin t \\ \downarrow \\ \sin t &= \frac{x}{a} \end{aligned}$$





用三角换元法

$$\begin{array}{c} a \\ \text{---} \\ t \\ \text{---} \\ \sqrt{a^2-x^2} \end{array} \quad \begin{array}{l} \sin t = \frac{x}{a} \\ \cos t = \frac{\sqrt{a^2-x^2}}{a} \end{array}$$

$$\begin{aligned} \therefore \text{原式} &= \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{a^2}{2} \frac{x}{a} \cdot \frac{\sqrt{a^2-x^2}}{a} + C \\ &= \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{x}{2} \sqrt{a^2-x^2} + C \end{aligned}$$

例 $\int \frac{1}{\sqrt{x^2+a^2}} dx \quad (a>0)$

令 $x = a \tan t \quad t \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$\begin{aligned} \text{原式} &= \int \frac{1}{\sqrt{a^2(1+\tan^2 t)}} d(a \tan t) \\ &= \int \frac{a \sec^2 t}{a \cdot \sec t} dt \\ &= \int \sec t dt \end{aligned}$$

由(21) $= \ln |\sec t + \tan t| + C_1$

$$\tan t = \frac{x}{a} \quad \frac{\sqrt{a^2+x^2}}{a} \quad \sec t = \frac{\sqrt{a^2+x^2}}{a}$$

$$\begin{aligned} \text{原式} &= \ln \left| \frac{\sqrt{a^2+x^2}}{a} + \frac{x}{a} \right| + C_1 \\ &= \ln |\sqrt{x^2+a^2} + x| - \ln |a| + C_1 \\ &= \ln (x + \sqrt{x^2+a^2}) + C \end{aligned}$$

(27) $\int \frac{1}{\sqrt{x^2+a^2}} dx = \ln(x + \sqrt{x^2+a^2}) + C$

同理 (28) $\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln |x + \sqrt{x^2-a^2}| + C$

例: $\int \frac{1}{\sqrt{x} + \sqrt[6]{x}} dx$ 令 $\sqrt[6]{x} = t \quad x = t^6$

$$\begin{aligned} \text{原式} &= \int \frac{6t^5 dt}{t^3 + t^2} \\ &= 6 \int \frac{t^3 dt}{t+1} \\ &= 6 \int \frac{t^3+1-1}{t+1} dt \\ &= 6 \int (t^2-t+1 - \frac{1}{t+1}) dt \\ &= 6(\frac{1}{3}t^3 - \frac{1}{2}t^2 + t - \ln(1+t)) + C \\ &= 2\sqrt[6]{x} - 3\sqrt[3]{x} - 6\ln(1+\sqrt[6]{x}) + C \end{aligned}$$

例: $\int x^2(1-x)^{1000} dx$ 令 $t=1-x \quad x=1-t$

$$\begin{aligned} \text{原式} &= \int (1-t)^2 t^{1000} d(1-t) \\ &= - \int (1-2t+t^2) t^{1000} dt \\ &= - \int (t^{1000} - 2t^{1001} + t^{1002}) dt \\ &= - \frac{1}{1001} t^{1001} + \frac{2}{1002} t^{1002} - \frac{1}{1003} t^{1003} + C \\ &= - \frac{1}{1001} (1-x)^{1001} + \frac{1}{501} (1-x)^{1002} - \frac{1}{1003} (1-x)^{1003} + C \end{aligned}$$





$$\int \frac{x^2}{(2x+1)^{10}} dx$$

令 $2x+1=t, x=\frac{t-1}{2}$

$$\text{原式} = \int \frac{(t-1)^2}{4t^{10}} \cdot \frac{1}{2} dt$$

$$= \frac{1}{8} \int \frac{t^2 - 2t + 1}{t^{10}} dt$$

第54讲 不定积分的分部积分.

例: $\int x e^x dx$

定理: 若 $u(x), v(x)$ 可导, 则

$$\int u dv = uv - \int v du$$

证: 由 $(uv)' = \cancel{uv'} + \cancel{u'v} + u'v + uv'$

$$\Rightarrow uv' = (uv)' - u'v$$

$$\Rightarrow \int uv' dx = \int [(uv)' - u'v] dx$$

$$\Rightarrow \int u dv = \int (uv)' dx - \int u'v dx$$

$$\int u dv = uv - \int v du$$

证法二: $d(uv) = v du + u dv$

$$\Rightarrow u dv = d(uv) - v du$$

$$\int u dv = \int d(uv) - \int v du$$

$$\int u dv = uv - \int v du$$

称为分部积分

$v'(x)$ 原函数易求.

求 $\int f(x) dx = \int u(x) v'(x) dx$

$$= \int u dv = uv - \int v du$$

$$= u(x) v(x) - \int v(x) u'(x) dx.$$

例: $\int x e^x dx$

解: 令 $u=x, v'=e^x \Rightarrow v=e^x$

$$\text{原式} = \int \frac{x}{u} \frac{de^x}{dv}$$

$$= \int x \frac{de^x}{dv}$$

$$= x e^x - \int e^x dx$$

$$= (x-1) e^x + C$$

若选错了, 则

$$\int x e^x dx = \int e^x d(\frac{1}{2} x^2)$$

$$= \frac{1}{2} e^x \cdot x^2 - \int \frac{1}{2} x^2 de^x$$

$$= \frac{1}{2} e^x \cdot x^2 - \int \frac{1}{2} x^2 e^x dx \text{ 更麻烦}$$

设 $P_k(x)$ 是 x 的 k 次多项式, $a \neq 0$ 常数.

$$1. \int \frac{P_k(x)}{u} \frac{e^{ax}}{v'} dx = \int P_k(x) d(\frac{1}{a} e^{ax})$$

需 k 次不定积分

$$2. \int P_k(x) \cos ax dx = \int P_k(x) d(\frac{\sin ax}{a})$$

$$3. \int P_k(x) \sin ax dx = \int P_k(x) d(-\frac{\cos ax}{a})$$





设 $p(x)$ 为 x 的函数.

$$4. \int p(x) f(\arcsin x) dx$$

能凑则凑, 如果不能凑

$$\text{令 } f(\arcsin x) = u \quad p(x) = v'$$

求出 v , 使用分部积分.

$$5. \int p(x) f(\ln x) dx$$

能凑则凑, 若不能凑.

$$\text{令 } f(\ln x) = u \quad p(x) = v'$$

求出 v , 使用分部积分.

$$\text{例: } \int (1+x^2) \cos 2x dx$$

$$= \int (1+x^2) d(\frac{1}{2} \sin 2x)$$

$$= \frac{1}{2} \sin 2x (1+x^2) - \int (\frac{1}{2} \sin 2x) d(1+x^2)$$

$$= \frac{1}{2} \sin 2x (1+x^2) - \int x \sin 2x dx$$

$$= \frac{1}{2} \sin 2x (1+x^2) + \int x d(\frac{1}{2} \cos 2x)$$

$$= \frac{1}{2} (1+x^2) \sin 2x + \frac{x}{2} \cos 2x - \frac{1}{2} \int \cos 2x dx$$

$$= \frac{1}{2} (1+x^2) \sin 2x + \frac{1}{2} x \cos 2x - \frac{1}{4} \sin 2x + C.$$

$$\int \frac{\arctan x}{1+x^2} dx$$

$$= \int \arctan x d \arctan x$$

$$= \frac{1}{2} (\arctan x)^2 + C$$

$$\text{例: } \int \frac{\arctan x}{u} \frac{dx}{dv}$$

$$= x \arctan x - \int x \frac{1}{1+x^2} dx$$

$$= x \arctan x - \frac{1}{2} \int \frac{1}{1+x^2} d(x^2+1)$$

$$= x \arctan x - \frac{1}{2} \ln(1+x^2) + C$$

$$\text{例: } \int \frac{\arctan x}{x^2(1+x^2)} dx$$

$$= \int (\frac{1}{x^2} - \frac{1}{1+x^2}) \arctan x dx$$

$$= -\frac{1}{2} (\arctan x)^2 + \int \arctan x d(-\frac{1}{x})$$

$$= -\frac{1}{2} (\arctan x)^2 - \frac{1}{x} \arctan x + \int (\frac{1}{x}) d \arctan x$$

$$= -\frac{1}{2} (\arctan x)^2 - \frac{1}{x} \arctan x + \int \frac{1}{x(1+x^2)} dx$$

$$\text{由 } \int \frac{1}{x} \cdot \frac{1}{1+x^2} dx = \int (\frac{1}{x} - \frac{x}{1+x^2}) dx$$

$$= \ln|x| - \frac{1}{2} \ln(1+x^2) + C$$

$$\therefore \text{原式} = -\frac{1}{2} (\arctan x)^2 + \frac{1}{x} \arctan x$$

$$- \ln|x| + \frac{1}{2} \ln(1+x^2) + C$$

$$\text{例: } \int \frac{\ln^2 x}{x} dx$$

$$= \int \ln^2 x d \ln x$$

$$= \frac{1}{3} \ln^3 x + C$$

$$\text{例: } \int \sqrt{x} \ln x dx$$

$$= \int \ln x d(\frac{2}{3} x^{\frac{3}{2}})$$

$$= \frac{2}{3} x^{\frac{3}{2}} \ln x - \int \frac{2}{3} x^{\frac{3}{2}} \cdot \frac{1}{x} dx$$

$$= \frac{2}{3} x^{\frac{3}{2}} \ln x - \frac{4}{9} x^{\frac{3}{2}} + C$$





例: 求 $\int \frac{e^{ax}}{v'} \frac{u}{u} dx$ ($a \neq 0, b \neq 0$ 常)

$$= \int \cos bx d\left(\frac{1}{a} e^{ax}\right)$$

$$= \frac{1}{a} e^{ax} \cos bx - \frac{1}{a} \int e^{ax} d \cos bx$$

$$= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \int \frac{e^{ax}}{v'} \frac{\sin bx}{u} dx$$

$$= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \int \sin bx d\left(\frac{1}{a} e^{ax}\right)$$

$$= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b}{a^2} \int e^{ax} d \sin bx$$

$$= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b^2}{a^2} \int e^{ax} \cos bx dx$$

设 $I = \int e^{ax} \cos bx dx$, 有

$$I = \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b^2}{a^2} I$$

$$a^2 I = a e^{ax} \cos bx + b e^{ax} \sin bx - b^2 I$$

$$I = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

同理可求 $\int e^{ax} \sin bx dx$.

例: $\int \frac{\sqrt{a^2 - x^2}}{u} \frac{dv}{dv} dx$ ($a > 0$).

$$= x \sqrt{a^2 - x^2} - \int x d \sqrt{a^2 - x^2}$$

$$= x \sqrt{a^2 - x^2} - \int \frac{-2x^2}{2\sqrt{a^2 - x^2}} dx$$

$$= x \sqrt{a^2 - x^2} - \int \frac{-x^2}{\sqrt{a^2 - x^2}} dx$$

$$= x \sqrt{a^2 - x^2} - \int \frac{a^2 - x^2 - a^2}{\sqrt{a^2 - x^2}} dx$$

$$\begin{aligned} \text{原式} &= x \sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} dx + \\ &+ a^2 \arcsin \frac{x}{a} \end{aligned}$$

$$= \int \sqrt{a^2 - x^2} dx = x \sqrt{a^2 - x^2} + a^2 \arcsin \frac{x}{a}$$

$$\therefore \int \sqrt{a^2 - x^2} dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{a} + C$$

第55讲 有理函数、三角函数的有理式
无理函数的不定积分

有理函数的不定积分

设 $P_n(x)$, $Q_m(x)$ 分别是 n -次与 m -次的多项式, 称 $\frac{P_n(x)}{Q_m(x)}$ 为有理函数.

当 $n \geq m$ 时称为有理假分式函数.

当 $n < m$ 时称为有理真分式函数.

利用多项式除法一定可以把有理假分式化成多项式与有理真分式之和.

$$\frac{x^4 + 3x^2 + x + 1}{x^2 + x + 1} = x^2 - x + 3 + \frac{-x - 2}{x^2 + x + 1}$$

$$\begin{array}{r} x^2 + x + 1 \overline{) x^4 + 0 \cdot x^3 + 3x^2 + x + 1} \\ \underline{x^4 + x^3 + x^2} \\ -x^3 + 2x^2 + x \\ \underline{-x^3 - x^2 - x} \\ 3x^2 + 2x + 1 \\ \underline{3x^2 + 3x + 3} \\ -x - 2 \end{array}$$

$$-x^3 + 2x^2 + x$$

$$-x^3 - x^2 - x$$

$$3x^2 + 2x + 1$$

$$3x^2 + 3x + 3$$

$$-x - 2$$





$$\frac{x^2-x+5}{(x+1)(x-2)^2(x^2-x+1)(x^2+x+1)^2}$$

$$= \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2} + \frac{D}{(x-2)^3}$$

$$+ \frac{Ex+F}{x^2-x+1} + \frac{Gx+H}{x^2+x+1} + \frac{Ix+J}{(x^2+x+1)^2}$$

通分. 整理. 系数对应相同.

例: $\frac{x^3+x-1}{(x-1)^2(x^2-x+1)^3}$

$$= \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2-x+1} + \frac{Ex+F}{(x^2-x+1)^2} + \frac{Gx+H}{(x^2-x+1)^3}$$

$$\int \frac{A}{(x-a)^n} dx \quad \int \frac{Ax+B}{(x^2+px+q)^n} dx$$

第一类最简分式. 第二类最简分式.

当 $n=1$ $\int \frac{A}{x-a} dx$

$$= A \int \frac{1}{x-a} dx - a$$

$$= A \ln|x-a| + C$$

当 $n>1$. $\int \frac{A}{(x-a)^n} dx$

$$= A \int (x-a)^{-n} d(x-a)$$

$$= \frac{A}{1-n} \frac{1}{(x-a)^{n-1}} + C$$

求 $I_n = \int \frac{1}{(x^2+a^2)^n} dx$ ($n \in \mathbb{N}$, $a \neq 0$ 常数)

解: $I_n = \frac{1}{a^2} \int \frac{a^2+x^2-x^2}{(x^2+a^2)^n} dx$ 当 $n>1$ 时

$$= \frac{1}{a^2} \int \frac{1}{(x^2+a^2)^{n-1}} dx - \frac{1}{a^2} \int \frac{x \cdot \frac{x}{(x^2+a^2)^n} dx}{\frac{u}{v'}}$$

$$\left(\begin{aligned} \text{令 } v' &= \frac{x}{(x^2+a^2)^n} \quad v = \int \frac{x}{(x^2+a^2)^n} dx \\ \text{原式} &= \int (x^2+a^2)^{-n} x dx = \frac{1}{2} \int (x^2+a^2)^{-n} d(x^2+a^2) \\ &= \frac{1}{-2(n-1)} (x^2+a^2)^{1-n} = -\frac{1}{2(n-1)} \frac{1}{(x^2+a^2)^{n-1}} \end{aligned} \right)$$

$$I_n = \frac{1}{a^2} I_{n-1} + \frac{1}{2a^2(n-1)} \int x d \frac{1}{(x^2+a^2)^{n-1}}$$

$$= \frac{1}{a^2} I_{n-1} + \frac{1}{2a^2(n-1)} \cdot \frac{x}{(x^2+a^2)^{n-1}}$$

$$- \frac{1}{2a^2(n-1)} \int \frac{1}{(x^2+a^2)^{n-1}} dx \quad I_{n-1}$$

$$= \frac{1}{a^2} \frac{2n-3}{2(n-1)} I_{n-1} + \frac{1}{2a^2(n-1)} \cdot \frac{x}{(x^2+a^2)^{n-1}}$$

$$I_1 = \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$$

逐渐从 n 到 1.

$$\int \frac{Ax+B}{(x^2+px+q)^n} dx \quad (p^2-4q < 0, \text{不能再分解})$$

解: 若 $A \neq 0$ $\int \frac{Ax+B}{(x^2+px+q)^n} dx$

$$= \frac{A}{2} \int \frac{2x + \frac{2B}{A}}{(x^2+px+q)^n} dx$$





$$d(x^2+px+q)=(2x+p)dx$$

$$\text{原式} = \frac{A}{2} \int \frac{2x + \frac{2B}{A} + p - p}{(x^2+px+q)^n} dx$$

$$= \frac{A}{2} \int \frac{2x+p}{(x^2+px+q)^n} dx + (B - \frac{Ap}{2}) \int \frac{dx}{(x^2+px+q)^n}$$

$$= \frac{A}{2} \int \frac{1}{(x^2+px+q)^n} d(x^2+px+q)$$

$$+ (B - \frac{Ap}{2}) \int \frac{1}{[(x+\frac{p}{2})^2 + (\frac{4q-p^2}{4})]^n} d(x+\frac{p}{2})$$

$$\left(\begin{aligned} q - \frac{p^2}{4} &= \frac{4q-p^2}{4} \quad \because p^2-4q < 0 \quad \therefore 4q-p^2 > 0 \\ &= \left(\frac{\sqrt{4q-p^2}}{2} \right)^2 = a^2 \end{aligned} \right)$$

前半部分由上上例解决. 后半部分由上例解决.

$$\text{例: } \int \frac{1}{x^4-1} dx$$

$$= \int \frac{1}{(x^2-1)(x^2+1)} dx$$

$$= \frac{1}{2} \int \left(\frac{1}{x^2-1} - \frac{1}{x^2+1} \right) dx$$

$$= -\frac{1}{2} \int \left(\frac{1}{1-x^2} + \frac{1}{1+x^2} \right) dx$$

$$= -\frac{1}{2} \left[\frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + \arctan x \right] + C$$

$$\text{解法: } \frac{1}{x^4-1} = \frac{1}{(x+1)(x-1)(x^2+1)}$$

$$= \frac{A}{x+1} + \frac{B}{x-1} + \frac{Cx+D}{x^2+1}$$

$$\text{赋值法. } 1 = A(x-1)(x^2+1) + B(x+1)(x^2+1) + (Cx+D)(x+1)(x-1)$$

由多项式相等成立知 一切 x 代入均成立

令 $x=-1$

$$1 = -4A \quad A = -\frac{1}{4}$$

令 $x=1$

$$1 = 4B \quad B = \frac{1}{4}$$

令 $x=0$

$$1 = -A + B - D \quad D = -\frac{1}{2}$$

令 $x=2$

$$1 = 5A + 15B + (2C - \frac{1}{2}) \times 3 \quad C=0$$

$$\text{例: } \int \frac{x^4+2x^3+2x+1}{x^2+x+1} dx$$

$$= \int \left(x^2+x-2 + \frac{3x+3}{x^2+x+1} \right) dx$$

$$= \frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x + 3 \int \frac{x+1}{x^2+x+1} dx$$

$$\int \frac{x+1}{x^2+x+1} dx = 2x^2 \frac{1}{2} \int \frac{2x+2}{x^2+x+1} dx$$

$$= \frac{1}{2} \int \frac{2x+1+1}{x^2+x+1} dx$$

$$= \frac{1}{2} \int \frac{d(x^2+x+1)}{x^2+x+1} + \frac{1}{2} \int \frac{d(x+\frac{1}{2})}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}$$

$$= \frac{1}{2} \ln(x^2+x+1) \quad \text{用公式}$$

$$+ \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \arctan \frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}} + C$$





例: $\int \frac{x^4+1}{x^6+1} dx$

$$= \int \frac{x^4+1-x^2+x^2}{(x^2+1)(x^4-x^2+1)} dx$$

$$= \int \frac{1}{x^2+1} dx + \int \frac{x^2}{x^4-x^2+1} dx$$

$$= \arctan x + \frac{1}{3} \int \frac{1}{(x^3)^2+1} d(x^3)$$

$$= \arctan x + \frac{1}{3} \arctan(x^3) + C$$

例: $\int \frac{x^2+1}{x^4+1} dx$

$$= \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} dx$$

$$= \int \frac{1}{x^2+\frac{1}{x^2}-2+2} d(x-\frac{1}{x})$$

$$= \int \frac{1}{(x-\frac{1}{x})^2+(\sqrt{2})^2} d(x-\frac{1}{x})$$

$$= \frac{1}{\sqrt{2}} \arctan \frac{x-\frac{1}{x}}{\sqrt{2}} + C$$

求 $\int R(\sin x, \cos x) dx$

令 $\tan \frac{x}{2} = t$

$$\cos x = 2 \cos^2 \frac{x}{2} - 1$$

$$= \frac{2}{\sec^2 \frac{x}{2}} - 1$$

$$= \frac{2}{1+\tan^2 \frac{x}{2}} - 1$$

$$= \frac{1-\tan^2 \frac{x}{2}}{1+\tan^2 \frac{x}{2}}$$

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$= \frac{2 \sin \frac{x}{2}}{\cos \frac{x}{2}} \cos^2 \frac{x}{2}$$

$$= \frac{2 \tan \frac{x}{2}}{1+\tan^2 \frac{x}{2}}$$

$$\tan \frac{x}{2} = t \quad \frac{x}{2} = \arctan t \quad x = 2 \arctan t$$

$$dx = \frac{2}{1+t^2} dt$$

$$\int R(\sin x, \cos x) dx$$

$$= \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2}{1+t^2} dt$$

转化为有理函数的不定积分.

定理: 有理函数的不定积分一定存在.

结果是多项式或有理函数
或对数函数或三角函数.

第56讲 三角函数有理式的
不定积分.

由 $\sin x, \cos x, \tan x, \sec x, \cot x$

$\csc x$ 及常数经过四则运算

得到的表达式记为

$$R(\sin x, \cos x, \tan x, \cot x, \sec x, \csc x)$$

称为三角函数的有理式.

$$= R(\sin x, \cos x).$$





1. $\int R(\sin^2 x, \sin x \cos x, \cos^2 x) dx$

$$\sin^2 x = \frac{\sin^2 x}{\cos^2 x} \cdot \cos^2 x = \frac{\tan^2 x}{1 + \tan^2 x}$$

$$\sin x \cos x = \frac{\sin x}{\cos x} \cos^2 x = \frac{\tan x}{1 + \tan^2 x}$$

$$\cos^2 x = \frac{1}{\sec^2 x} = \frac{1}{1 + \tan^2 x}$$

原式 = $\int R(\tan x) dx$

$$= \int R(\tan x) \frac{\sec^2 x}{\sec^2 x} dx$$

$$= \int R(\tan x) \frac{1}{1 + \tan^2 x} d \tan x.$$

能凑则凑, 如果不能凑,

令 $\tan x = t$.

$$\text{原式} = \int R(t) \frac{1}{1+t^2} dt$$

例: $\int \frac{1}{\sin x \cos^3 x} dx$

解: 原式 = $\int \frac{\sec^2 x}{\sin x \cos x} dx$

$$= \int \frac{1}{\frac{\sin x}{\cos x} \cdot \cos^2 x} d \tan x$$

$$= \int \frac{1 + \tan^2 x}{\tan x} d \tan x$$

$$= \ln |\tan x| + \frac{1}{2} \tan^2 x + C$$

解法 = : 原式 = $\int \frac{\sin^2 x + \cos^2 x}{\sin x \cos^3 x} dx$

$$= \int \frac{\sin x}{\cos^3 x} dx + \int \frac{1}{\sin x \cos x} dx$$

$$= -\int \frac{1}{\cos^3 x} d \cos x + \int \frac{1}{\frac{1}{2} \sin 2x} dx$$

$$= \frac{1}{2} \sec^2 x + \int \csc 2x d(2x)$$

$$= \frac{1}{2} \sec^2 x + \ln |\csc 2x - \cot 2x| + C$$

2. $\int \sin^n x \cos^m x dx$.

(1) n, m 中至少有一个为奇数, 另一个指数可以是任意的常数.

此时把奇数分离出一次幂, 用凑微分.

$$\int \sin^{\frac{1}{2}} x \cos^3 x dx$$

$$= \int \sin^{\frac{1}{2}} x \cos^2 x \cos x dx$$

$$= \int \sin^{\frac{1}{2}} x (1 - \sin^2 x) d \sin x$$

$$= \int (\sin^{\frac{1}{2}} x - \sin^{\frac{5}{2}} x) d \sin x$$

$$= \frac{4}{5} \sin^{\frac{5}{2}} x - \frac{4}{13} \sin^{\frac{13}{2}} x + C$$

(2) m, n 均为偶数, 利用三角公式降幂, 或者用线性运算法则或用 (1)





例 $\int \cos^2 x \sin^2 x dx$ $\sin^2 x \cos^2 x = (\frac{1}{2} \sin 2x)^2$

解: 原式 = $\int \frac{1}{2} (1 + \cos 2x) \cdot \frac{1}{4} \sin^2 2x dx$
 $= \frac{1}{8} \int \sin^2 2x dx + \frac{1}{8} \int \cos 2x \sin^2 2x dx$
 $= \frac{1}{16} \int (1 - \cos 4x) dx + \frac{1}{16} \int \sin^2 2x d(\sin 2x)$
 $= \frac{1}{16} (x - \frac{1}{4} \sin 4x + \frac{1}{3} \sin^3 2x) + C$

3. $\int \sin 4x \cos 3x dx$

解: 原式 = $\frac{1}{2} \int (\sin 7x + \sin x) dx$
 $= \frac{1}{2} (-\frac{1}{7} \cos 7x - \cos x) + C$

第57讲 无理函数的不定积分

1. $\int R(\sqrt[n]{\frac{ax+b}{cx+d}}, x) dx$

令 $\sqrt[n]{\frac{ax+b}{cx+d}} = t$ 解出 $x = \varphi(t)$ 有理式

原式 = $\int R(t, \varphi(t)) \varphi'(t) dt$

$\sqrt[n]{ax+b} = t$

$\sqrt{(x-a)(x-b)} = |x-a| \sqrt{\frac{x-b}{x-a}}$

令 $\sqrt{\frac{x-b}{x-a}} = t$

2. $\int R(x, \sqrt{ax^2+bx+c}) dx, (a \neq 0)$

首先把根式化成下列形式之一.

$\sqrt{q^2 - \varphi^2(x)}$ 或 $\sqrt{q^2 + \varphi^2(x)}$ 或 $\sqrt{\varphi^2(x) - q^2}$

其中 $\varphi(x) = hx + p$ ($h \neq 0$) q 为常数.

$dx = \frac{1}{h} d(hx+p) = \frac{1}{h} d\varphi(x)$

在心中把 $\varphi(x)$ 看成 u . 能凑则凑, 如果不能凑, 就要去根式.

① 令 $\varphi(x) = q \cdot \sin t$. $d\varphi(x) = q \cdot \cos t dt$

$hx+p = q \cdot \sin t$ $x = \frac{1}{h}(q \sin t - p)$

② 令 $\varphi(x) = q \tan t$

③ 令 $\varphi(x) = q \sec t$

例: $\int \frac{1}{\sqrt{x^2+x+1}} dx$

$= \int \frac{1}{\sqrt{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}} d(x+\frac{1}{2})$

$= \ln(x+\frac{1}{2} + \sqrt{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}) + C$

$= \ln(x+\frac{1}{2} + \sqrt{x^2+x+1}) + C$



